ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q3, Q4

Question 1

Given: sensor 1280 px wide, pixel pitch p = 5 µm (square), lens f = 16 mm, standoff Z = 400 mm; force sensor 10 mV/N, 12-bit ADC over 0–5 V; plant $\dot{x} = -2x + u$ with zero-order hold, T = 0.1 s.

(a) (i) $f_x = f / p = 16\text{ mm} / 5\ \mu\text{m} = 16\times10^{-3} / 5\times10^{-6} =$ **3200 px**.

(ii) Sensor width $W_s = 1280 \times 5\ \mu\text{m} = 6.4$ mm. By similar triangles FOV $= W_s \cdot Z / f = 6.4 \times 400 / 16 =$ **160 mm**.

Distance per pixel = FOV / 1280 = 160/1280 = **0.125 mm/px** (check: $Z/f_x = 400/3200 = 0.125$ mm/px).

(b) LSB $= 5\text{ V}/2^{12} = 5/4096 = 1.2207\text{ mV} = 1.2207\times10^{-3}$ V.

$\Delta F = \text{LSB}/S$ with S = 10 mV/N = 10 V/N (taking the mV as V in the denominator): $\Delta F = 1.2207\times10^{-3}/10 =$ **1.22×10^{-4} N**.

(c) $F = e^{AT} = e^{-0.2} = 0.8187$. $G = A^{-1}(F - I)B = (0.8187 - 1)/(-2) = 0.0906$.

Hence the exact model is $x_{k+1} =$ **$0.0906\ x_k + 0.8187\ u_k$**.

Question 3

Given: A = [0, 1; 0, −2], B = [0; 1], C = [1, 0], f_s = 1 kHz so the sample time is $T = 1/f_s$ = 1 ms.

(a) (1) Differentiation is a high-pass operation (gain ∝ ω), so encoder measurment/quantisaton noise is amplified, by about 1/T = 1000 at 1 kHz.

(2) Encoder quantisation: Δθ is a whole number of counts, so ω̂ = Δθ/T moves in steps of (count size)/T; at low speed the estimate is a coarse staircase of zeros and spikes.

(b) The analogue prototype is the first-order low-pass $H(s) = \omega_c/(s + \omega_c)$ with $\omega_c = 2\pi f_c = 2\pi \times 20 = 125.66$ rad/s. The bilinear (Tustin) transform replaces s by $(2/T)(z - 1)/(z + 1)$ with $T = 1/f_s = 0.001$ s.

Define $a = \omega_c T/2 = 125.66 \times 0.001/2 = 0.062832$. Then $H(z) = \omega_c / [(2/T)(z-1)/(z+1) + \omega_c] = a(z + 1) / [(1 + a)\ z - (1 - a)] = a(1 + z^{-1}) / [(1 + a) - (1 - a)\ z^{-1}]$.

Dividing through by (1 + a): $a/(1 + a) = 0.062832/1.062832 = 0.059117$ and $(1 - a)/(1 + a) = 0.937168/1.062832 = 0.881767$. Hence with u = differenced signal and y = filtered velocity: **$y_k = 0.05912\ (u_k + u_{k-1}) + 0.88177\ y_{k-1}$**. (Frequency pre-warping is neglected since $f_c \ll f_s/2$.)

Verification of the DC gain: set u and y constant (z = 1): $y = 2(0.05912)u + 0.88177\ y \Rightarrow y(1 - 0.88177) = 0.11824\ u \Rightarrow y/u = 0.11824/0.11823 = 1.000$. In general $H(1) = 2a/[(1 + a) - (1 - a)] = 2a/2a = 1$ exactly, so the steady-state output equals the commanded input ✓.

(c) $A - LC = [-\ell_1, 1; -\ell_2, -2]$. $\det(sI - A + LC) = (s + \ell_1)(s + 2) + \ell_2 = s^2 + (2 + \ell_1)s + (2\ell_1 + \ell_2)$.

Desired: $(s + 10)^2 = s^2 + 20s + 100$. Match: $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$; $2\ell_1 + \ell_2 = 100 \Rightarrow \ell_2 = 64$. **$L = [18, 64]^T$**.

(d) Advantage of the observer: it uses the plant model and the applied input u, so its velocity estimate has (almost) no phase lag; the 20 Hz filter on differenced data always delays by about $1/(2\pi f_c) \approx 8$ ms. It can also reconstruct states that cannot be obtained by differencing (e.g. disturbance, when augmented).

Filter is safer when the model is uncertain or changes (unknown payload, friction, contact, actuator saturation): the obsever then gives a biased or poorly-behaved estimate while the model-free fliter still follows the measured data.

Sampling at 1 kHz is fast compared with the 10 rad/s obsever bandwidth, so discretisation effects are small.

Question 4

Given: plant $\ddot{\theta} + a\dot{\theta} + c\theta = b\,u$ with $a = 2$, $c = 6$, $b = 1$; controller $u = N_{ff}\,r - k_1\theta - k_2\dot{\theta}$.

(a) Closed loop: $\ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\,r$, i.e. $s^2 + 4s + 8 = 0$.

$s = [-4 \pm \sqrt{(16 - 32)}]/2 = -4/2 \pm \sqrt{(-16)} = \mathbf{-2 \pm 4j}$.

DC gain $= N_{ff}/8 = 1$, so $\mathbf{N_{ff} = 8}$.

(b) With $b = 0.75$ ($N_{ff} = 8$, $k_1 = 2$): $G_{DC} = bN_{ff}/(c + bk_1) = 0.75 \times 8/(6 + 0.75 \times 2) = 6/7.5 = 0.8$.

$\theta_{ss} = \mathbf{0.8}$. Tracking error $= (1 - 0.8)/1 \times 100 = \mathbf{20\%}$. (Less than the 25% actuator loss, because the stiffness also dropped.)

(c) Substituting u and differentiating: the integral term enters with a minus sign, so the characteristic polynomial is $s^3 + (a + bk_2)s^2 + (c + bk_1)s - bK_I$.

Desired: $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

So $2 + k_2 = 8 \Rightarrow k_2 = 6$; $6 + k_1 = 24 \Rightarrow k_1 = 18$; $-K_I = 32 \Rightarrow \mathbf{K_I = -32}$.

(d) For a constant r, in stedy state all derivatives vanish: $\dot{\xi} = r - \theta_{ss} = 0 \Rightarrow \theta_{ss} = r$, independent of b and c; the closed loop only needs to be stable so that a steady state exists.

Failure mode: integrator windup. When the actuator saturates the error persists, ξ keeps growing and causes a large overshoot afterwards. Remedy: anti-windup (clamp the integrator, conditional integration or back-calculation).